Notes on a 2-adic observable for the Collatz map, with a speculative arithmetic-topology dictionary

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These are public research notes, not a paper. They record (i) an empirical study of a scalar observable $H(\gamma_n)$ defined on finite Collatz trajectories, (ii) a tentative dictionary relating features of the Collatz dynamics on $\mathbb{Z}_2$ to entries in the M^2KR arithmetic-topology dictionary and to objects in Kim's arithmetic Chern–Simons theory, and (iii) an empirical observation about the support-size threshold $|S_k| \leq 4$ that came up during this study. Nothing here is claimed as a theorem. The dictionary is offered as a speculative organizing structure that may or may not survive contact with the actual mathematical objects involved; the empirical results stand on their own and are reproducible from the accompanying probe.

Familiarity with the Collatz map and basic p-adic notation is assumed. No background in arithmetic topology is required: the relevant ideas are used only as heuristic motivation.

The numbers reported here come from `collatz_arithmetic_probe.py`, available at the repository above under CC0.

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1 Background and motivation

Several independent strands feed into what follows.

Mazur's 1964 observation [1] that prime ideals in Spec($\mathbb{Z}$) behave formally like knots in a 3-manifold has been developed into what is now called arithmetic topology, with the dictionary worked out systematically by Morishita [5], Kapranov [2], and Reznikov [3]; we refer to the resulting correspondence table as M^2KR. Kim's arithmetic Chern–Simons theory [6] provides the cohomological machinery used to attach holonomy and linking-type invariants to paths in arithmetic fundamental groups. Siegel's (p, q)-adic spectral framework [9] establishes that the shortened Collatz map is conjugate to the 2-adic shift on $\mathbb{Z}_2$, which is what allows Collatz trajectories to be discussed as paths in a p-adic setting at all. Orzechowski's recent theorem [10] on minimal paradoxical decompositions of non-Archimedean spaces fixes the number four as the minimum piece-count for such a decomposition. The empirical 2-adic transfer-operator framework on which the observable is defined was set up in earlier notes [11].

What these notes do is record a few measurements made when these ingredients are placed next to one another, together with the speculative dictionary that suggested the measurements in the first place. None of the dictionary entries are proven correspondences; they are offered as targets for verification. The empirical numbers are what they are.

2 Setup on $\mathbb{Z}_2$

2.1 The shortened map

Throughout, T_3 denotes the shortened Collatz map

$$T_3(n) = \begin{cases} n/2 & n \text{ even}, \\ (3n+1)/2 & n \text{ odd}. \end{cases}$$

A finite trajectory starting at n is the sequence $\gamma_n = (n, T_3(n), T_3^2(n), \ldots, T_3^{|\gamma_n|}(n))$, terminating when the iterate enters the attractor cycle $\{1, 2\}$.

For $N \geq 1$, set $S_0 = \{1, \ldots, N\}$ and let μ_0 be the uniform measure on S_0. The pushforward

$$\mu_{k+1} = (T_3)_* \mu_k, \qquad S_{k+1} = T_3(S_k)$$

defines the measure evolution. The observable used throughout is the 2-adic absolute value $f(n) = |n|_2 = 2^{-v_2(n)}$, where $v_2(n) = \max\{j \geq 0 : 2^j \mid n\}$, with expectation

$$\mathbb{E}_{\mu_k}[f] = \frac{1}{|S_k|} \sum_{n \in S_k} 2^{-v_2(n)}.$$

2.2 Supermartingale conjecture

Conjecture 1 (Supermartingale property, [11])**.** *$\mathbb{E}_{\mu_k}[f]$ is a supermartingale up to bounded fluctuations with respect to the natural filtration $\mathcal{F}_k = \sigma(S_0, \ldots, S_k)$:*

$$\mathbb{E}\big[\mathbb{E}_{\mu_{k+1}}[f] \,\big|\, \mathcal{F}_k\big] \leq \mathbb{E}_{\mu_k}[f] + \varepsilon_k,$$

with $\varepsilon_k \geq 0$, ε_k bounded, and $\sum_k \varepsilon_k < \infty$ in expectation under the ensemble measure.

The empirical sweep in Section 5 is consistent with this in its Cesàro form across six scales of N. Step-by-step monotonicity is not asserted and is in fact violated at every scale tested.

2.3 The four-piece threshold

Following [11], the loss of structural richness in the ensemble as μ_k collapses toward the attractor is tracked by five measures $M_1, \ldots, M_5$. The relevant landmark here is the first step at which the support shrinks to four elements, $|S_k| \leq 4$. The number four is singled out because Orzechowski [10] shows that any field with a nontrivial non-Archimedean valuation admits a paradoxical decomposition into exactly four pieces under its affine isometry group, and four is the minimum. No mechanism relating these phenomena is known. Whether the coincidence of this minimum with the support size at which the Collatz ensemble loses statistical structure is meaningful or coincidental is open; the empirical behaviour at the crossing is reported in Section 5. Note that the image-diversity measure $M_2 = |T_3(S_k)|$ at the same crossing step is typically 3 rather than 4, so "four-piece crossing" here refers to support size, not M_2.

2.4 The 2-adic conjugacy

Siegel [9] establishes that T_3 is topologically and measurably conjugate to the 2-adic shift σ on $\mathbb{Z}_2$. The conjugacy does not preserve finitely supported measures: in particular, it does not preserve

collapse times or the distributional properties of finite ensembles, since the natural embedding $\mathbb{N} \hookrightarrow \mathbb{Z}_2$ is not respected by the conjugating homeomorphism. Empirically, collapse times for T_3 on $\{1, \ldots, N\}$ over $N \in \{50, \ldots, 2000\}$ grow markedly slower than any polynomial rate (69, 74, 78, 90, 112, 114 steps across the sweep), with the corresponding orbits of σ on truncated 2-adic ensembles collapsing substantially faster; the precise asymptotic form in this range is not claimed here. We use the conjugacy in the following specific sense: (i) it identifies T_3-orbits with σ-orbits as sequences in $\mathbb{Z}_2$, and (ii) it preserves the 2-adic topology and the 2-adic valuation v_2. Neither of these requires preservation of finite-support measures, and only these two properties are used in what follows. This identification places Collatz trajectories in $\mathbb{Z}_2 = \varprojlim \mathbb{Z}/2^n\mathbb{Z}$, which is the setting in which the speculative dictionary below is well-formed enough to write down.

3 The M^{2}KR dictionary and Kim's theory: very brief recap

The foundational analogy:

3-dimensional topology	**Algebraic number theory**
Knot $K \subset S^3$	Prime $\mathfrak{p} \subset \mathrm{Spec}(\mathbb{Z})$
3-manifold M	Ring of integers $\mathcal{O}_K$
Knot complement $S^3 \setminus K$	$\mathrm{Spec}(\mathbb{Z}) \setminus \{\mathfrak{p}\}$
Linking number	Legendre symbol
Alexander polynomial	Iwasawa polynomial
Milnor $\bar{\mu}$-invariants	Multiple residue symbols
Knot group $\pi_1(S^3 \setminus K)$	Galois group with restricted ramification

Of the three layers of formalization here, only the Alexander/Iwasawa correspondence is currently a theorem; Kim's program is the active formalization of the second layer.

Kim's framework [6] constructs a Chern–Simons-type functional in the arithmetic setting, with $\mathrm{Spec}(\mathcal{O}_K)$ in the role of the 3-manifold and p-adic Galois representations in the role of gauge connections. The central object is the unipotent fundamental group $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}) \setminus S, b)$ for a basepoint b and a finite set of primes S. Paths in this space carry holonomy data in a p-adic Lie group. Subsequent work [8, 7] connects this machinery to p-adic L-functions and to entanglement-type quantities in the abelian case.

4 A speculative dictionary

A Collatz trajectory $(n, T_3(n), T_3^2(n), \ldots)$ is a sequence of positive integers and, via $\mathbb{N} \hookrightarrow \mathbb{Z}_2$, a sequence in the 2-adic tree. Whether such sequences should be regarded as "paths" in any sense useful to Kim's theory is exactly what is at issue. The table below proposes a row-by-row correspondence; it is offered as an organizing scheme, not as a list of established facts.

The following table should be read strictly as a hypothesis generator: several rows are likely incorrect or incomplete.

Verification status applies to the Collatz column specifically: **(V)** indicates the topology–arithmetic correspondence is directly verified against Morishita [4] Section 1 *and* the Collatz identification is essentially tautological; **(V/P)** indicates the topology–arithmetic correspondence is verified but the Collatz identification is proposed and interpretive; **(P)** indicates a row original to these notes.

Collatz	**Topology**	**Arithmetic**	
Trajectory $\gamma_n = \{T_3^k(n)\}$ as sequence in $\mathbb{Z}_2$	Knot $K \hookrightarrow S^3$, $S^1 \hookrightarrow M$	Prime $\mathrm{Spec}(\mathbb{F}_\mathfrak{p}) \hookrightarrow \mathrm{Spec}(\mathcal{O}_k)$	**(V/P)**
Ambient ring $\mathbb{Z}_2$	3-manifold $M \simeq \mathrm{Spec}(\mathcal{O}_k)$	$\mathrm{Spec}(\mathcal{O}_k)$	**(V)**
Collatz support S_k shrinking toward attractor	Knot complement $X_K = M \setminus V_K^\circ$	$\mathrm{Spec}(\mathcal{O}_k) \setminus \{\mathfrak{p}\}$	**(V/P)**
Parity branching $B(n) = \mathbf{1}_{n \text{ odd}}$	Meridian $\alpha \in \pi_1(\partial V_K)$	Inertia generator $\tau \in I_\mathfrak{p}$	**(V/P)**
T_3-step direction (even/odd choice)	Longitude $\beta \in \pi_1(\partial V_K)$	Frobenius $\sigma \in \pi_1(\mathrm{Spec}(\mathbb{F}_\mathfrak{p}))$	**(V/P)**
Monodromy of Collatz sequences in $\hat{\mathbb{Z}}$	Knot group $G_K = \pi_1(S^3 \setminus K)$	Galois group G_S with restricted ramification	**(V/P)**
v_2-step under T_3	Linking number $\mathrm{lk}(K, L)$	Legendre symbol $\left(\frac{p}{q}\right)$	**(V/P)**
Char. polynomial of v_2-transfer; supermartingale generating function	Alexander polynomial $\Delta_K(t)$	Iwasawa polynomial $f_K(T)$	**(V/P)**
Multi-step v_2 correlations	Milnor $\bar{\mu}$-invariants	Multiple residue symbols / Massey products	**(V/P)**
Attractor $\{1, 2\}$	Unknot; $G_K = \mathbb{Z}$, $\Delta_K = 1$	Unramified extension; $\pi_1(\mathrm{Spec}(\mathbb{Z})) = 1$	**(V/P)**
Threshold $\lvert S_k \rvert = 4$	Minimal 4-piece non-Arch. paradoxical decomposition	(speculative) ramification boundary	**(P)**
Transfer operator $\mu \mapsto (T_3)_* \mu$	Deck transformation; covering monodromy	Iwasawa-algebra action on class group	**(P)**

Remark 1. *The fundamental topology–arithmetic correspondences ($S^1 \leftrightarrow \mathrm{Spec}(\mathbb{F}_q)$, knot $\leftrightarrow$ prime, meridian $\leftrightarrow$ inertia τ, longitude $\leftrightarrow$ Frobenius σ) appear as Morishita's eq. (1.2) [4]; the Legendre/linking and Alexander/Iwasawa correspondences are Sections 2 and 7 of the same survey. The proposed identification of $B(n)$ with inertia rather than Frobenius is interpretive: $B(n)$ heuristically tracks whether the trajectory passes through the ramified (odd) branch, while the step direction tracks Frobenius. Whether either identification holds up under closer scrutiny is open.*

4.1 The observable $H(\gamma_n)$

Definition 1 (Trajectory-summed 2-adic depth)**.** *For a finite Collatz trajectory* $\gamma_n = (n, T_3(n), \ldots, T_3^{|\gamma_n|}(n))$ *terminating at the attractor, with* $|\gamma_n|$ *the number of steps, set*

$$H(\gamma_n) = \sum_{j=0}^{|\gamma_n|-1} v_2(\gamma_n^{(j)}) \in \mathbb{Z}_{\geq 0},$$
$$H_{\text{norm}}(\gamma_n) = \frac{H(\gamma_n)}{|\gamma_n|} \in \mathbb{Q}_{\geq 0},$$
$$H_{\log}(\gamma_n) = H(\gamma_n) \cdot \log 2 \in \mathbb{R}_{\geq 0}.$$

The sum in H *runs over all nodes of the trajectory except the terminal one.*

If the dictionary in Section 4 survives, $H(\gamma_n)$ is a candidate scalar shadow of an underlying $\mathbb{Z}_p$-valued holonomy in the abelianized setting — i.e., the kind of object Kim's theory would attach to a path. The lift to that framework would require specifying a basepoint in $\pi_1^{\text{un}}(\text{Spec}(\mathbb{Z}_2))$ and a target p-adic Lie group, neither of which is done here. For now H is simply a number assigned to each trajectory, and what follows is a record of its empirical behaviour.

5 Empirical results

All numbers in this section are produced by `collatz_arithmetic_probe.py` [11].

5.1 Supermartingale sweep

Across $N \in \{50, 100, 200, 500, 1000, 2000\}$:

N	steps	$\sum_k \varepsilon_k$	violations	max ratio	min ratio	mean ratio
50	69	2.637	31	1.500	0.714	1.012
100	74	2.248	36	1.440	0.788	1.009
200	78	1.394	36	1.447	0.788	1.005
500	90	1.687	42	1.333	0.679	1.007
1000	112	2.651	53	1.482	0.583	1.008
2000	114	1.255	61	1.286	0.804	1.003

The Cesàro mean C_k is non-increasing across all tested N. Upward step-ratio violations exist at every scale, but the mean ratio is uniformly within $[1.003, 1.012]$ of unity, and $\sum_k \varepsilon_k < 2.7$ throughout, well below $\mathbb{E}_{\mu_0}[f] \approx 0.5$–$0.7$. The proximity of the mean ratio to 1 across all tested scales is the most stable numerical feature observed in this study. This is consistent with Conjecture 1 in its Cesàro form. Whether the normalization remains stable as $N \to \infty$ is open empirically.

5.2 H on small trajectories

$H(\gamma_n)$ was computed for $n = 1, \ldots, 100$. Observed regularities:

- *Approximately linear in length:* Pearson correlation of H with trajectory length is $r = 0.959$. H is sensitive to properties of the trajectory as a whole rather than to the starting point alone.
- *Weak dependence on* $v_2(n)$*:* correlation of H with $v_2(n)$ is $r = -0.243$. The starting 2-adic depth has limited predictive power for the trajectory total.

- *Normalized mean:* the mean of H_{norm} over $n \leq 100$ is 1.22. For comparison, if the 2-adic digits along trajectories were uniformly distributed, the per-step mean of v_2 would be $\sum_{k\geq 0} k \cdot 2^{-(k+1)} = 1$; the observed excess of 0.22 reflects the non-uniform digit distribution actually produced by the iteration.
- *Parity:* $H \equiv v_2(n) \pmod 2$ in 55% of cases, not distinguishable from chance at sample size 100; recorded for completeness rather than as a structural observation.

Selected values:

n	$v_2(n)$	length	H	H_{norm}
1	0	1	0	0.000
2	1	1	1	1.000
8	3	3	6	2.000
16	4	4	10	2.500
27	0	70	47	0.671

The famously long trajectory at $n = 27$ has the lowest normalized H in the sample, consistent with approximate linearity in length rather than dependence on the arithmetic depth of the start point.

5.3 H near the attractor

The attractor cycle $\{1, 2\}$ has signature

$$v_2 \text{ sequence: } [0, 1, 0], \qquad H = 1, \qquad H_{\text{norm}} = \tfrac{1}{3}.$$

The ratio $2^{v_2(2)}/2^{v_2(1)} = 2$ matches the signature $M_4 = 2$ from [11].

Tail H (last K steps before the attractor, computed for $n = 3, \ldots, 500$):

K	mean	std	min	max
5	1.500	0.395	1.200	3.000
10	1.424	0.593	1.000	4.500
20	1.185	0.180	0.900	2.650

The tail H tightens with window size: at $K = 20$ the standard deviation is 0.18, substantially narrower than the full-trajectory distribution ($\sigma = 0.61$). The asymptotic tail value sits in a narrow band around ≈ 1.18 rather than at a single universal number.

5.4 Behaviour at the four-piece crossing

For $N = 200$ (78 collapse steps), the threshold $|S_k| \leq 4$ is first crossed at step 75, i.e. at 96.2% of total collapse time. State at the crossing:

Quantity	Value at crossing
Support size $\lvert S_k \rvert$	4
Support set	$\{1, 2, 4, 5\}$
$\bar{v}_2$	0.750
σ_{v_2}	0.829
M_1 (conditional gap)	0.625
M_2 (image diversity)	3
M_3 (branch balance)	1.000
M_4 ($\mathbb{E}[f \lvert B{=}0]/\mathbb{E}[f \lvert B{=}1]$)	0.375
M_5 (between-branch variance)	0.098

The v_2-mean drops sharply at the crossing: pre-crossing window mean 1.424, post-crossing window mean 0.944, $\Delta\bar{v}_2 = -0.480$ (symmetric five-step windows on either side). The mean continues to fall toward the attractor value 0.5.

So the empirical picture at $N = 200$ is: the trajectory spends roughly 96% of its lifetime above $|S_k| = 4$, then crosses the threshold and collapses to the attractor within the remaining $\sim 4\%$ of steps, with a measurable drop in $\bar{v}_2$ at the crossing. Whether this reflects something structural about the four-piece minimum or is a generic feature of late-trajectory collapse is not settled by the data; it does suggest the crossing is a useful landmark to measure against.

6 Open questions

The questions below are what would need to be resolved to turn any of the above into something more than empirical observation plus a speculative dictionary. Tagged *[analytic]* for those needing pen-and-paper proof, *[structural]* for those needing precise identification of an object in an existing framework.

Q1. *[structural]* **Verification of the dictionary.** For each row of the table in Section 4, write down a precise statement of the proposed correspondence using Morishita [5] and Kim [6], and identify rows that need modification or should be dropped. The parity-versus-inertia identification is the most fragile.

Q2. *[structural]* **Lifting H.** Specify (i) the basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$, (ii) the target p-adic Lie group, and (iii) the sense in which discrete Collatz steps define morphisms in the étale fundamental groupoid, in such a way that the resulting holonomy reduces to H in the abelian quotient and is consistent with $H_{\mathrm{norm}}(\{1, 2\}) = 1/3$ and the tail value $\approx 1.18 \pm 0.18$.

Q3. *[analytic]* **Proof of the supermartingale conjecture.** The Cesàro mean is empirically non-increasing across $N \in \{50, \ldots, 2000\}$ with $\sum_k \varepsilon_k < 2.7$ throughout. A proof is a prerequisite for any optional-stopping argument applied to H.

Q4. *[structural]* **Status of $H_{\mathrm{norm}} = 1/3$.** Is the attractor value 1/3 canonical in any meaningful sense, or just an artefact of the cycle length?

Q5. *[structural]* **The four-piece coincidence.** Is the support-size-4 crossing actually related to Orzechowski's four-piece minimum, or is it coincidence? If related, what (if anything) does it correspond to in Kim's setting?

Q6. *[structural]* **Non-abelian extension.** Whether the non-abelian version of Kim's theory is accessible via Collatz trajectories at all.

Q7. *[analytic]* **Reverse implication.** Whether well-definedness and convergence of a lifted H would imply Collatz, or a conditional version of it, via optional stopping.

7 Summary

What these notes contain: a scalar observable H defined on finite Collatz trajectories; an empirical record of its behaviour ($r = 0.96$ correlation with trajectory length, normalized mean 1.22, tail value 1.18 ± 0.18, attractor value $1/3$); a supermartingale signature for $\mathbb{E}_{\mu_k}[f]$ that is consistent across six scales of N; an observation that the four-piece threshold $|S_k| \leq 4$ is a sharply localized late-trajectory event (96.2% of total time at $N = 200$) coinciding with a $\Delta\bar{v}_2 = -0.48$ drop; and a speculative dictionary that motivated looking at H in the first place.

What these notes do not contain: any proven correspondence with arithmetic topology or Kim's theory, any proof of the supermartingale conjecture, or any claim that the four-piece coincidence is more than a coincidence. The dictionary is offered as a hypothesis to be tested against the actual mathematical objects, not as a result.

References

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